

Amanuel Fesseha

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EDUCATION

Georgia State University

Associates of Science, Electrical Engineering

Atlanta, GA

May 2024

University of Georgia

Bachelor of Science, Electrical Engineering

Athens, GA

May 2027

Cumulative GPA: 3.7/4.0

RELEVANT PROJECT EXPERIENCE

CURO Research Fellowship, *Biomedically Adaptive Leaping System*

Summer 2025

- Engineered a bio-inspired leaping robot in Fusion 360 with focus on energy efficiency and actuator performance.
- Built custom brushless motor actuators with FOC control, delivering ~18 Nm torque.
- Constructed URDF/Xacro for ROS2 Jazzy and Gazebo simulation, resolving mesh and transform issues.
- Optimized gear ratios and control pipelines to improve locomotion adaptability.
- Researched kangaroo-inspired mechanics to enhance energy efficiency and terrain adaptability.
- Presented findings in technical forums, highlighting results, limitations, and future linkage designs.

Create Lab Project, *Projects*

September 2024 - Present

- Reaction Latency Evaluation** - Developed a modern Reaction Latency System using ESP32 and industrial-grade LED strips to deliver precise, reliable timing measurements for gerontology research.
- PPG measuring device** - Developed a compact PPG device to measure heart rate and blood oxygen levels, enabling portable and efficient biometric monitoring.
- Pecan Impact Recording Device** - Built a pecan-sized wireless impact-sensing pod using an nRF52840 MCU with a 6-axis IMU, implementing high-rate sampling and pre/post-impact buffering with onboard logging. Designed schematic/PCB, shock-resistant packaging, and validated low-power 2.4 GHz/BLE data transfer.

RELEVANT WORK EXPERIENCE

Create Lab

Athens, GA

Student Engineer

September 2024 - Present

- Engineered custom products with cross-functional teams, translating client briefs into manufacturable designs.
- Applied engineering principles and problem-solving techniques to create solutions tailored to client specifications.
- Managed project timelines, ensuring efficient delivery of high-quality products within specified deadlines.
- Conducted product testing and provided technical feedback to optimize performance and functionality.
- Worked closely with clients to refine project requirements and align final product designs with their expectations.

Georgia State University

Atlanta, GA

Student Administrative Assistant

September 2021 - August 2024

- Streamlined procurement and invoicing to boost administrative throughput.
- Accurately and promptly handled the submission of travel expense reports ensuring process efficiency.
- Coordinated with suppliers for timely delivery, ensuring operational continuity.
- Processed faculty/staff reimbursements with audit-ready accuracy and policy compliance.

INVOLVEMENT EXPERIENCE

UGA Robotics Club, *Member* | Athens, GA

August 2024 - Present

- Designed and built small-scale combat robots, integrating mechanical and electrical systems with lightweight components fabricated using additive manufacturing techniques. Conducted iterative testing and optimization to enhance performance and durability, utilizing engineering tools to get a competitive edge.

TECHNICAL SKILLS

- Languages:** Python, C/C++, SQL; Tools: Git/GitHub, Jupyter: Java, Python, C
- Embedded:** ESP32 (S3/C3), nRF52840; I2C, SPI, UART; BLE and integration of Sensors and IO
- Software:** Altium Designer, KiCad (schematic/PCB), Fusion 360
- Robotics and ML:** IsaacSim, ROS2, Gazebo, URDF/Xacro; PyTorch CNNs and transfer learning